

Q1.

The time, in weeks, that a patient must wait to be given an appointment in Holmsoon Hospital may be modelled by a random variable T having the cumulative distribution function

$$F(t) = \begin{cases} 0 & t < 0 \\ \frac{t^3}{216} & 0 \leq t \leq 6 \\ 1 & t > 6 \end{cases}$$

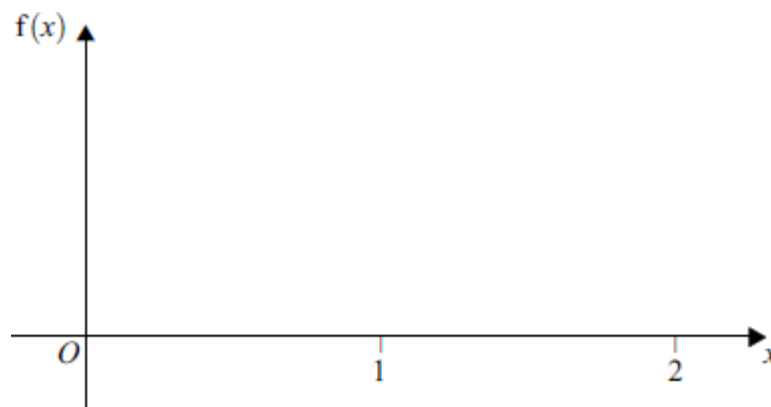
- (a) Find, to the nearest day, the time within which 90 per cent of patients will have been given an appointment. (3)
 - (b) Find the probability density function of T for all values of t . (3)
 - (c) Calculate the mean and the variance of T . (6)
 - (d) Calculate the probability that the time that a patient must wait to be given an appointment is more than one standard deviation above the mean. (4)
- (Total 16 marks)**

Q2.

A continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq 1 \\ \frac{1}{3}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f on the axes below.



- (b) (i) Find the cumulative distribution function, F , for $0 \leq x \leq 1$.

(ii) Hence, or otherwise, find the value of the lower quartile of X .

(2)

(c) (i) Show that the cumulative distribution function for $1 \leq x \leq 2$ is defined by

$$F(x) = \frac{1}{3} (5x - x^2 - 3)$$

(4)

(ii) Hence, or otherwise, find the value of the upper quartile of X .

(4)

(Total 15 marks)

Q3.

The continuous random variable X has a cumulative distribution function defined by

$$F(x) = \begin{cases} 0 & x < -5 \\ \frac{x+5}{20} & -5 \leq x \leq 15 \\ 1 & x > 15 \end{cases}$$

(a) Show that, for $-5 \leq x \leq 15$, the probability density function, $f(x)$, of X is given by

$$f(x) = \frac{1}{20}.$$

(1)

(b) Find:

(i) $P(X \geq 7)$;

(1)

(ii) $P(X \neq 7)$;

(1)

(iii) $E(X)$;

(1)

(iv) $E(3X^2)$.

(3)

(Total 7 marks)

Q4.

The continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}(x^2 + 1) & 0 \leq x \leq 1 \\ \frac{1}{4}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$F(x) =$

- (a) The cumulative distribution function of X is denoted by $F(x)$.

Show that, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{8}x(x^2 + 3)$$

(3)

- (b) Hence, or otherwise, verify that the median value of X is 1.

(2)

- (c) Show that the upper quartile, q , satisfies the equation $q^2 - 5q + 5 = 0$ and

hence that $q = \frac{1}{2}(5 - \sqrt{5})$.

(5)

- (d) Calculate the **exact** value of $P(q < X < 1.5)$.

(4)

(Total 14 marks)

Q5.

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The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{1}{8}(x - 4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State values for the median and the lower quartile of X .

(2)

- (b) Show that, for $1 \leq x \leq 4$, the cumulative distribution function, $F(x)$, of X is given by

$$F(x) = 1 + \frac{1}{54}(x - 4)^3$$

(You may assume that $\int (x-4)^2 dx = \frac{1}{3}(x-4)^3 + c$.)

(4)

(c) Determine $P(2 \leq X \leq 3)$.

(2)

(d) (i) Show that q , the upper quartile of X , satisfies the equation $(q-4)^3 = -13.5$.

(3)

(ii) Hence evaluate q to three decimal places.

(1)

(Total 12 marks)

Q6.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -c \\ \frac{x+c}{4c} & -c \leq x \leq 3c \\ 1 & x > 3c \end{cases}$$

where c is a positive constant.

(a) Determine $P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right)$.

(2)

(b) Show that the probability density function, $f(x)$, of X is

$$f(x) = \begin{cases} \frac{1}{4c} & -c \leq x \leq 3c \\ 0 & \text{otherwise} \end{cases}$$

(2)

(c) Hence, or otherwise, find expressions, in terms of c , for:

(i) $E(X)$;

(1)

(ii) $\text{Var}(X)$.

(1)

(Total 6 marks)

Q7.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} \frac{1}{16}x^3 & 0 \leq x \leq 2 \\ \frac{1}{6}(5-x) & 2 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Prove that the cumulative distribution function of X for $2 \leq x \leq 5$ can be written in the form

$$F(x) = 1 - \frac{1}{12}(5-x)^2$$

(4)

- (c) Hence, or otherwise, determine $P(X \geq 3 \mid X \leq 4)$.

(5)

(Total 12 marks)

Q8.

The waiting time, X minutes, for fans to gain entrance to see an event may be modelled by a continuous random variable having the distribution function defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leq x \leq 1 \\ \frac{1}{54}(x^3 - 12x^2 + 48x - 10) & 1 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

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- (a) (i) Sketch the graph of F .

(4)

- (ii) Explain why the value of q_1 , the lower quartile of X , is $\frac{1}{2}$.

(2)

- (iii) Show that the upper quartile, q_3 , satisfies $1.6 < q_3 < 1.7$.

(3)

- (b) The probability density function of X is defined by

$$f(x) = \begin{cases} \alpha & 0 \leq x \leq 1 \\ \beta(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Show that the **exact** values of α and β are $\frac{1}{2}$ and $\frac{1}{18}$ respectively.

(5)

- (ii) Hence calculate $E(X)$.

(5)

(Total 19 marks)

Q9.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$.

(2)

- (b) (i) Show that the probability density function of X is defined by

$$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

(2)

- (ii) Sketch the graph of f .

(2)

- (c) Given that $\text{Var}(X) = [E(X)]^2$, show that $k = 2 \pm \sqrt{3}$.

(5)

(Total 11 marks)

Q10.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$.

(2)

- (b) Determine an expression, in terms of k , for the lower quartile, q_1 .

(3)

- (c) Show that the probability density function of X is defined by

$$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

(2)

- (d) Given that $k = 11$:

- (i) sketch the graph of f ;

(2)

- (ii) determine $E(X)$ and $\text{Var}(X)$;

(2)

- (iii) show that $P(q_1 < X < E(X)) = 0.25$.

(2)

(Total 13 marks)

Q11.

The waiting time, T minutes, before being served at a local newsagents can be modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} \frac{3}{8}t^2 & 0 \leq t < 1 \\ \frac{1}{16}(t+5) & 1 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

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- (a) Sketch the graph of f .

(3)

- (b) For a customer selected at random, calculate $P(T \geq 1)$.

(2)

- (c) (i) Show that the cumulative distribution function for $1 \leq t \leq 3$ is given by

$$F(t) = \frac{1}{32}(t^2 + 10t - 7)$$

(5)

- (ii) Hence find the median waiting time.

(4)

(Total 14 marks)

Q12.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x \leq -4 \\ \frac{x+4}{9} & -4 \leq x \leq 5 \\ 1 & x \geq 5 \end{cases}$$

- (a) Determine the probability density function, $f(x)$, of X . (2)
- (b) Sketch the graph of f . (2)
- (c) Determine $P(X > 2)$. (2)
- (d) Evaluate the mean and variance of X . (2)

(Total 8 marks)

Q13.

Engineering work on the railway network causes an increase in the journey time of commuters travelling into work each morning.

The increase in journey time, T hours, is modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} 4t(1-t^2) & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $E(T) = \frac{8}{15}$. (3)
- (b) (i) Find the cumulative distribution function, $F(t)$, for $0 \leq t \leq 1$. (2)
- (ii) Hence, or otherwise, for a commuter selected at random, find

$$P(\text{mean} < T < \text{median})$$

(5)
(Total 10 marks)

Q14.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(2)

(b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x+1)$$

(3)

(ii) Hence find $P(X \geq 0.5)$.

(2)

(iii) Verify that the lower quartile, q_1 , satisfies

$$0.5 < q_1 < 0.75$$

(3)

(Total 10 marks)

Q15.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(2)

(b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x+1)$$

(3)

(ii) Hence write down the value of $P(X \leq 1)$.

(1)

(iii) Find the value of x for which $P(X \geq x) = \frac{17}{20}$.

(5)

(iv) Find the lower quartile of the distribution.

(4)
(Total 15 marks)

Q16.

Ann and Brian usually travel to work on the same train. The train is scheduled to leave the platform at 8.00 am, but it actually leaves at time T minutes after 8.00 am. The random variable T has an exponential distribution with distribution function:

$$F(t) = \begin{cases} 1 - e^{-\frac{t}{4}} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Write down the probability that the train leaves the platform at exactly 8.00 am. (1)
- (b) One morning Ann arrives on the platform at 8.02 am and Brian arrives at 8.06 am.
 - (i) Find the probability that the train has left before Ann arrives. (2)
 - (ii) Find the probability that Ann catches the train but Brian misses it. (3)
- (c) Another morning Brian arrives on the platform at 8.01 am and is in time to catch the train. Ann arrives at 8.04 am.
Find the probability that Ann also catches the train. (4)

(Total 10 marks)

Q17.

A continuous random variable T has the probability density function $f(t)$ given by:

$$f(t) = \begin{cases} \frac{1}{18}t^2 & 0 \leq t \leq 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch the graph of f . (2)
- (b) Determine the distribution function, $F(t)$, for $0 \leq t \leq 5$. (4)
- (c) Hence, or otherwise, calculate $P(T < 4)$. (2)

(Total 8 marks)

Q18.

The delay, T minutes, in connecting electronically to a ticket booking service can be modelled by the distribution function:

$$P(T \leq t) = 1 - e^{-2t}, \quad t \geq 0.$$

- (a) (i) Write down the probability density function of T . (1)
- (ii) State values for the mean and standard deviation of T . (2)
- (b) Calculate the probability that the delay exceeds 3 minutes. (2)
- (c) Find the median delay, giving your answer in seconds, to the nearest second. (3)

(Total 8 marks)



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